



Introduction to Blockchain

CSE-355

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Lecture 09

Tuesday, Feb 17, 2026

What Is a Token?

Simple Definition

A token is a digital unit of value recorded on a blockchain.

It is:

- Not a coin with its own blockchain
- But a programmable asset that lives **on top of** an existing blockchain

Coins vs Tokens

Coin

- Has its **own blockchain**
- Used to pay transaction fees
- Example: Bitcoin, Ether

Token

- Built on top of another blockchain
- Uses that blockchain for security
- Created via smart contracts

What Does a Token Represent?

A token can represent:

- Money (stablecoins)
- Voting rights
- Access rights
- Ownership shares
- Loyalty points
- Real-world assets

It's just:

A programmable entry in a distributed ledger.

How does a token work?

On blockchains that support smart contracts like Ethereum:

- A smart contract defines the token rules
- The ledger records balances
- Transfers are just state updates

Stablecoins are simply tokens that are designed to represent fiat money.

Stablecoins

- The alternative use case for blockchain
- Transactions involving stablecoins: on which blockchain?
- Must write transactions in a ledger (to have any record)

USDC(Circle)/USDT(Tether Limited)

A stablecoin is mapped to USD because the **issuer promises that 1 token = 1 USD**, and backs that promise with reserves.

Step 1 — Issuance

- You send \$1 to the issuer
- The issuer mints 1 stablecoin
- That \$1 is held in reserves (bank deposits, T-bills, etc.)

So now:

1 token is backed by \$1 held somewhere.

Step 2 — Redemption

- You return 1 stablecoin to the issuer
- The issuer burns (destroys) it
- You receive \$1

This **convertibility** keeps the price anchored.

Arbitrage

Why market price stays near \$1?

If the stablecoin trades:

- **Above \$1 (e.g., \$1.02)**
Traders buy from issuer for \$1 and sell for \$1.02 → price pushed down.
- **Below \$1 (e.g., \$0.98)**
Traders buy cheap and redeem for \$1 → price pushed up.
- This arbitrage keeps it close to \$1.

ARTs versus EMTs

Key Differences Between ARTs and EMTs

Feature	Asset-Referenced Tokens (ARTs)	Electronic Money Tokens (EMTs)
Pegging Mechanism	Multiple assets (fiat, commodities, crypto)	Single fiat currency (e.g., USD, EUR)
Use Case	Investment, store of value	Payments, transactions
Examples	Diem, tokenized gold	USDT, USDC, EUROe
Regulatory Requirement	Reserves must match assets referenced	Reserves must be in the same fiat currency
Issuers	Crypto firms, blockchain projects	Licensed Electronic Money Institutions (EMIs)

Bitcoin Transactions

Outline

- Introduction
- Replay attacks
- UTXO Model
- Bitcoin transaction format
- Stack based transaction execution
 - Legacy transactions
 - Transaction malleability
 - Segregated witness (segwit) transactions
- Merkle trees

Financial Transactions

What information needs to be included in a transaction?

- Payer
- Payee
- Amount
- Authorization

Payer: Alia

Payee: Babar

Amount: PKR 50,000

Authorization:

$\text{Sig}_{\text{Alia}}(\text{Payer} || \text{Payee} || \text{Amount})$

Financial Transactions

What information needs to be included in a transaction?

- Payer
- Payee
- Amount
- Authorization

Payer: Alia
Payee: Babar
Amount: PKR 50,000
Authorization:
 $\text{Sig}_{\text{Alia}}(\text{Hash}(\text{Payer} || \text{Payee} || \text{Amount}))$

Account-based model

Alia: PKR 100,000
Babar: PKR 0
Alia: PKR 50,000
Babar: PKR 50,000

Payer: Alia
Payee: Babar
Amount: PKR 50,000
Authorization:
 $\text{Sig}_{\text{Alia}}(\text{Hash}(\text{Payer} || \text{Payee} || \text{Amount}))$

Account-based model

- Store list of accounts and balances
- A transaction is valid if there is enough balance in the account
- Payer debited, payee credited

Ethereum uses account-based model

Replay Attack

Account-based model

Alia: PKR 100,000
Babar: PKR 0
Alia: PKR 50,000
Babar: PKR 50,000
Alia: PKR 0
Babar: PKR 100,000



Payer: Alia
Payee: Babar
Amount: PKR 50,000

Auth
Sig_{Alia}

Payer: Alia
Payee: Babar
Amount: PKR 50,000
Authorization:
Sig_{Alia}(Hash(Payer || Payee || Amount))

Replay Attacks

- Replay attacks can be prevented by associating unique Identifier (SEQ) with each transaction
 - This number is randomly generated whenever a transaction is generated
 - Payer signs, **$m = (\text{SEQ} || \text{Payer} || \text{Payee} || \text{Amount})$**
 - Can check if there are multiple entries of the same SEQ Number on the block chain
- Ethereum prevents replay attacks this way!!
 - Keeps track of the SEQ for each user and ensures that it is not reused
- Bitcoin uses a different model to prevent replay attacks or double spending
 - **Unspent Transaction Output (UTXO) model**

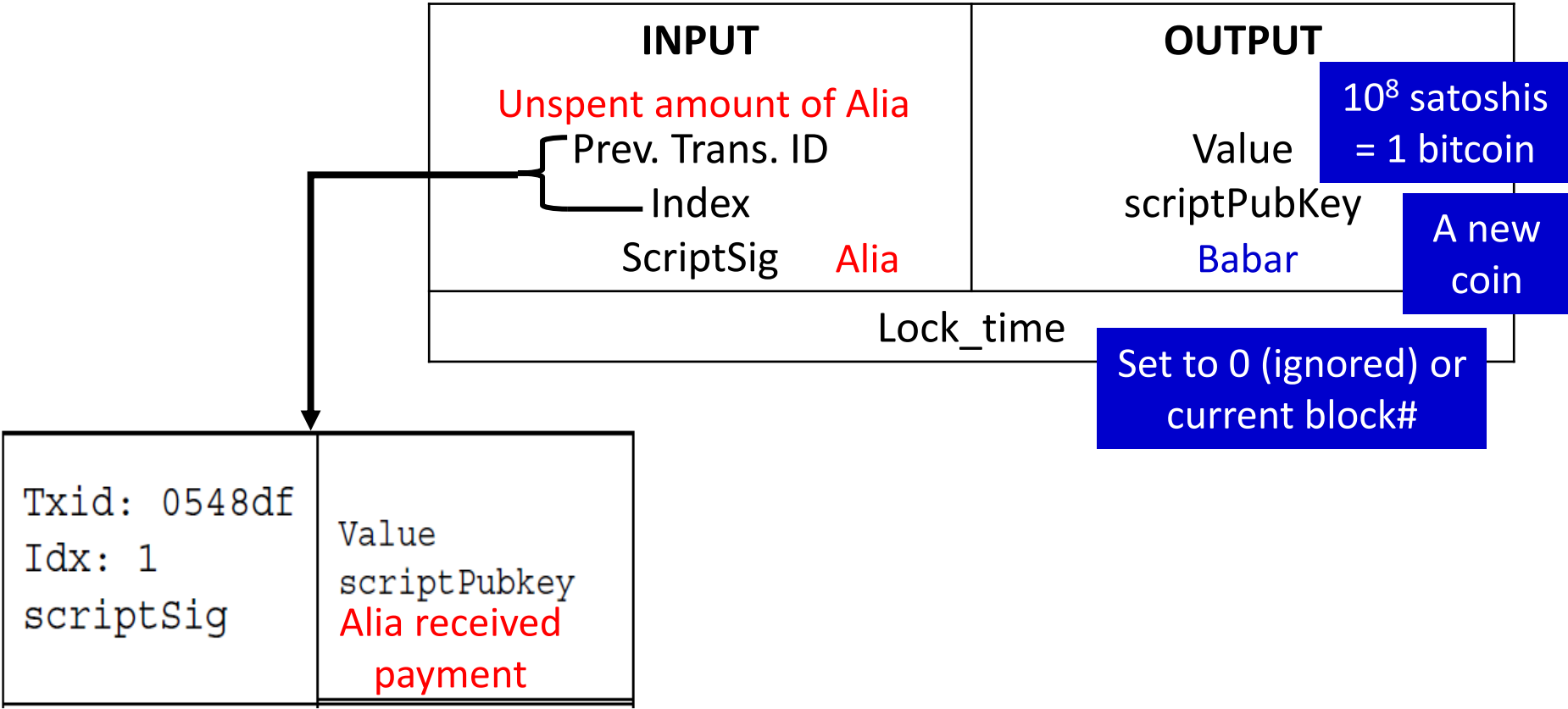
Unspent Transaction output (UTXO) Model

- A bitcoin can only be spent once
- Whenever a bit coin is spent, new coins are generated
- A transaction on Bitcoin blockchain consumes an unspent coin and generates new unspent coins

Bitcoin Transaction

Transaction Format (legacy)

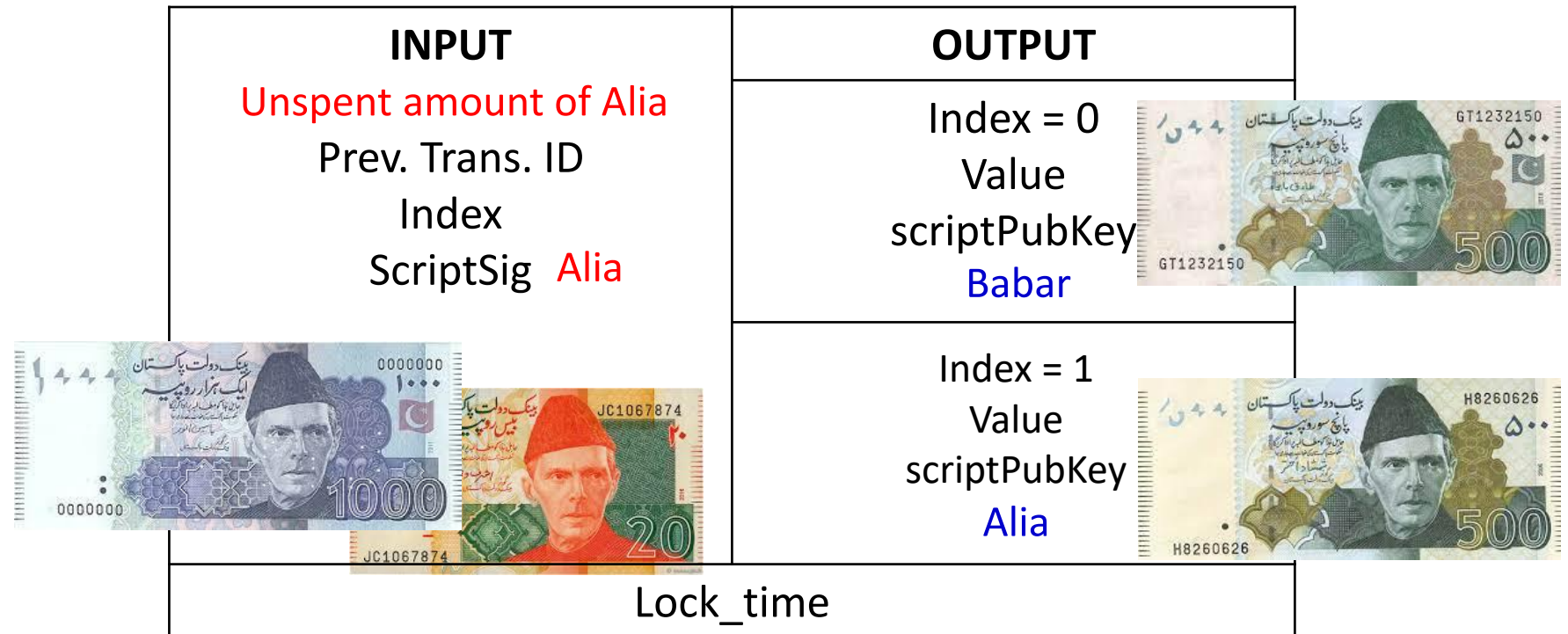
Alia paying to Babar



Bitcoin Transaction

Transaction Format (legacy)

Alia paying to Babar



<https://btcscan.org/>

ScriptSig and ScriptPubkey

ScriptSig

- Encodes the signature and public key of the payer

ScriptPubkey

- Sequence of instructions implementing a predicate (Boolean function)
- Takes ScriptSig as input
 - returns **True** if input corresponds to legitimate owner of the bitcoin, otherwise returns **False**

If ScriptPubKey(ScriptSig)
then create an output

Output – make payment to payee

lock payment to
Hash(public key of payee)
Pay to PubKey Hash

<https://btcscan.org/>

Lecture 10

Thursday, Feb 19, 2026

ScriptSig and ScriptPubkey

ScriptSig

- Encodes the signature and public key of the payer

```
<signature>
<public key>
```

ScriptPubkey

- Sequence of instructions implementing a predicate (Boolean function)
- Takes ScriptSig as input
 - returns **True** if input corresponds to legitimate owner of the bitcoin, otherwise returns **False**

```
OP_DUP
OP_HASH160
<H(pubkey)>
OP_EQUALVERIFY
OP_CHECKSIG
```

<https://btcscan.org/>

Bitcoin Transaction (legacy)

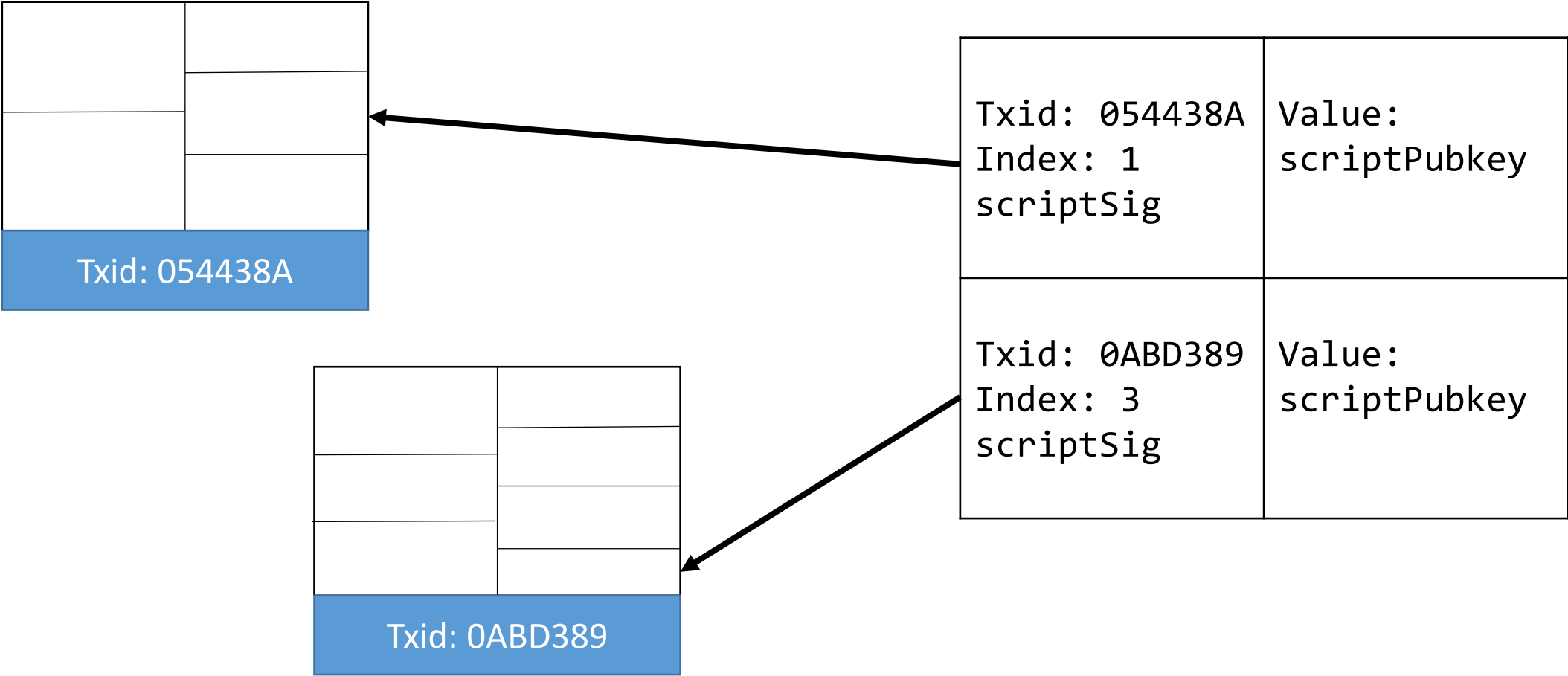
Pay to PubkeyHash (P2PKH)

- Payment is made to a public key
- Pubkeys are big in size (64 bytes), a hash of pubkey is only 20 bytes
- In the scriptPubKey, output (unspent bit coin) is locked to Hash of a Public key
- Only payee can unlock the unspent bit coin

```
OP_DUP
OP_HASH160
<H(pubkey)>
OP_EQUALVERIFY
OP_CHECKSIG
```

<https://btcscan.org/>

Bitcoin Transaction



Bitcoin Transaction Format (legacy)

```

"txid": "c80b343d2ce2b5d829c2de9854c7c8d423c0e33bda264c40138d834aaab4c0638",
"hash": "c80b343d2ce2b5d829c2de9854c7c8d423c0e33bda264c40138d834aaab4c0638",
"size": 225,
"version": 1,
"locktime": 0,
"vin": [
  {
    "txid": "3f4fa19803dec4d6a84fae3821da7ac7577080ef75451294e71f9b20e0ab1e7b",
    "vout": 0,
    "scriptSig": {
      "hex": "483045022100a1b2c3d4e5f60718293a4b5c6d7e8f901234567890abcdef1234567890ab..."
    },
    "sequence": 4294967295
  }
],
"vout": [
  {
    "value": 49.99990000,
    "n": 0,
    "scriptPubKey": {
      "asm": "OP_DUP OP_HASH160 cb c20a7664f2f69e5355aa427045bc15e7c6c772 OP_EQUALVERIFY OP_CHECKSIG",
      "reqSigs": 1,
      "type": "pubkeyhash",
      "addresses": ["mz6KvC4aoUeo6wSxtiVQTo7FDwPnkp6URG" ]
    }
  }
]

```



Bitcoin Transaction

Transaction ID:

[7cc90809ea6c90899086ed8607eea8dcc1bbe2ee3fbaa662f3f597888575c4c9](#)

INPUT	OUTPUT
Prev. Trans. ID: 0f07ec1c3738873cd6336f0e2c92094bf08d0ad3bce8aba0efa2eb9d172fd60c Index: 0 ScriptSig Alia 304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01 026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6a4304a5e680b2ab2cf8 Total Input: 0.00025302 + 0.01094000 + 0.00003616 = 0.01122918	Index 0 Value: 0.00025302 BTC scriptPubKey: OP_DUP Babar OP_HASH160 2fa9c4b9585b74fd5fd8c92fd4e32824b531e515 OP_EQUALVERIFY OP_CHECKSIG Index 1 Value: 0.01094000 BTC scriptPubKey: OP_DUP Alia OP_HASH160 9772ba84d827ac93d719a2a40f3f5dcb8963b8ad OP_EQUALVERIFY OP_CHECKSIG

Fees: 0.00003616 BTC

Execution

Transaction ID:

[7cc90809ea6c90899086ed8607eea8dcc1bbe2ee3fbba662f3f597888575c4c9](#)

INPUT	ScriptPubKey
Prev. Trans. ID: 0f07ec1c3738873cd6336f0e2c92094bf08d0ad3bce8aba0efa2eb9d172fd60c Index: 0 ScriptSig: Alia 304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01 026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6a4304a5e680b2ab2cf8	(from Prev. Transaction) OP_DUP OP_HASH160 <076960c3d0cdf8076ecd0cc220022384475aff76e> OP_EQUALVERIFY OP_CHECKSIG Value: 0.01122918 BTC

We need to check that the locked bitcoin in the previous transaction belong to Alia



Execution

Transaction ID:

[7cc90809ea6c90899086ed8607eea8dcc1bbe2ee3fbaa662f3f597888575c4c9](#)

ScriptSig:

<
304402205e4a6ac74d35de1ffc1188f717273bada586b3de
78f6844a548d548b0d27634c02207b127fa009b9d755b84
8772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01
> **Alia's Signatures**

<
026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6
a4304a5e680b2ab2cf8
> **Alia's public key**
 ScriptPubKey: (Prev. Trans.)
OP_DUP
OP_HASH160
<076960c3d0cdf8076ecd0cc2200223844
75aff76e>
OP_EQUALVERIFY
OP_CHECKSIG

Stack

Execution

Transaction ID:

[7cc90809ea6c90899086ed8607eea8dcc1bbe2ee3fbaa662f3f597888575c4c9](#)

ScriptSig:

<

304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01

> Alia's Signatures

<

026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6a4304a5e680b2ab2cf8

> Alia's public key

ScriptPubKey: (Prev. Trans.)

OP_DUP

OP_HASH160

<076960c3d0cdf8076ecd0cc220022384475aff76e>

OP_EQUALVERIFY

OP_CHECKSIG

Stack

304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01

Execution

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ScriptSig:

<

304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01

> **Alia's Signatures**

<

026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6a4304a5e680b2ab2cf8

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OP_CHECKSIG

Stack

026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6a4304a5e680b2ab2cf8

304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01

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ScriptSig:

```
<
304402205e4a6ac74d35de1ffc1188f717273bada586b3de
78f6844a548d548b0d27634c02207b127fa009b9d755b84
8772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01
> Alia's Signatures

<
026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6
a4304a5e680b2ab2cf8
> Alia's public key
    ScriptPubKey: (Prev. Trans.)
OP_DUP
OP_HASH160
<076960c3d0cdf8076ecd0cc2200223844
75aff76e>
OP_EQUALVERIFY
OP_CHECKSIG
```

Stack

026a7cd9a8f65b260bd459d8c1e823110a6993
91cbaabb6a4304a5e680b2ab2cf8

026a7cd9a8f65b260bd459d8c1e823110a6993
91cbaabb6a4304a5e680b2ab2cf8

304402205e4a6ac74d35de1ffc1188f717273ba
da586b3de78f6844a548d548b0d27634c02207
b127fa009b9d755b848772dcee21c4a0555450
1fd5e0d8a60ee8b9f9d13da7d01

Execution

Transaction ID:

[7cc90809ea6c90899086ed8607eea8dcc1bbe2ee3fbaa662f3f597888575c4c9](#)

ScriptSig:	Stack
< 304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01 > Alia's Signatures	
< 026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6a4304a5e680b2ab2cf8 > Alia's public key	026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6a4304a5e680b2ab2cf8
ScriptPubKey: (Prev. Trans.) OP_DUP OP_HASH160 <076960c3d0cdf8076ecd0cc220022384475aff76e> OP_EQUALVERIFY OP_CHECKSIG	026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6a4304a5e680b2ab2cf8
	304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01

Execution

Transaction ID:

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ScriptSig:	Stack
< 304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01 > Alia's Signatures	
< 026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6a4304a5e680b2ab2cf8 > Alia's public key	76960c3d0cdf8076ecd0cc220022384475aff76e
ScriptPubKey: (Prev. Trans.) OP_DUP OP_HASH160 <076960c3d0cdf8076ecd0cc220022384475aff76e> OP_EQUALVERIFY OP_CHECKSIG	026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6a4304a5e680b2ab2cf8
	304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01

Execution

Transaction ID:

[7cc90809ea6c90899086ed8607eea8dcc1bbe2ee3fbaa662f3f597888575c4c9](#)

ScriptSig:	Stack
< 304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01 > Alia's Signatures	76960c3d0cdf8076ecd0cc220022384475aff76e
< 026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6a4304a5e680b2ab2cf8 > Alia's public key	76960c3d0cdf8076ecd0cc220022384475aff76e
ScriptPubKey: (Prev. Trans.) OP_DUP OP_HASH160 <076960c3d0cdf8076ecd0cc220022384475aff76e> OP_EQUALVERIFY OP_CHECKSIG	026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6a4304a5e680b2ab2cf8
	304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01

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ScriptSig:

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304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01

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> **Alia's public key**

ScriptPubKey: (Prev. Trans.)

OP_DUP

OP_HASH160

<076960c3d0cdf8076ecd0cc220022384475aff76e>

OP_EQUALVERIFY

OP_CHECKSIG

Stack

76960c3d0cdf8076ecd0cc220022384475aff76e

76960c3d0cdf8076ecd0cc220022384475aff76e

026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6a4304a5e680b2ab2cf8

304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01

Execution

Transaction ID:

[7cc90809ea6c90899086ed8607eea8dcc1bbe2ee3fbaa662f3f597888575c4c9](#)

ScriptSig:

```
<
304402205e4a6ac74d35de1ffc1188f717273bada586b3de
78f6844a548d548b0d27634c02207b127fa009b9d755b84
8772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01
> Alia's Signatures

<
026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6
a4304a5e680b2ab2cf8
> Alia's public key
    ScriptPubKey: (Prev. Trans.)
OP_DUP
OP_HASH160
<076960c3d0cdf8076ecd0cc2200223844
75aff76e>
OP_EQUALVERIFY
OP_CHECKSIG
```

Stack

026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6a4304a5e680b2ab2cf8
304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01

Execution

Transaction ID:

[7cc90809ea6c90899086ed8607eea8dcc1bbe2ee3fbaa662f3f597888575c4c9](#)

ScriptSig:

<

304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01

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> **Alia's public key**

ScriptPubKey: (Prev. Trans.)

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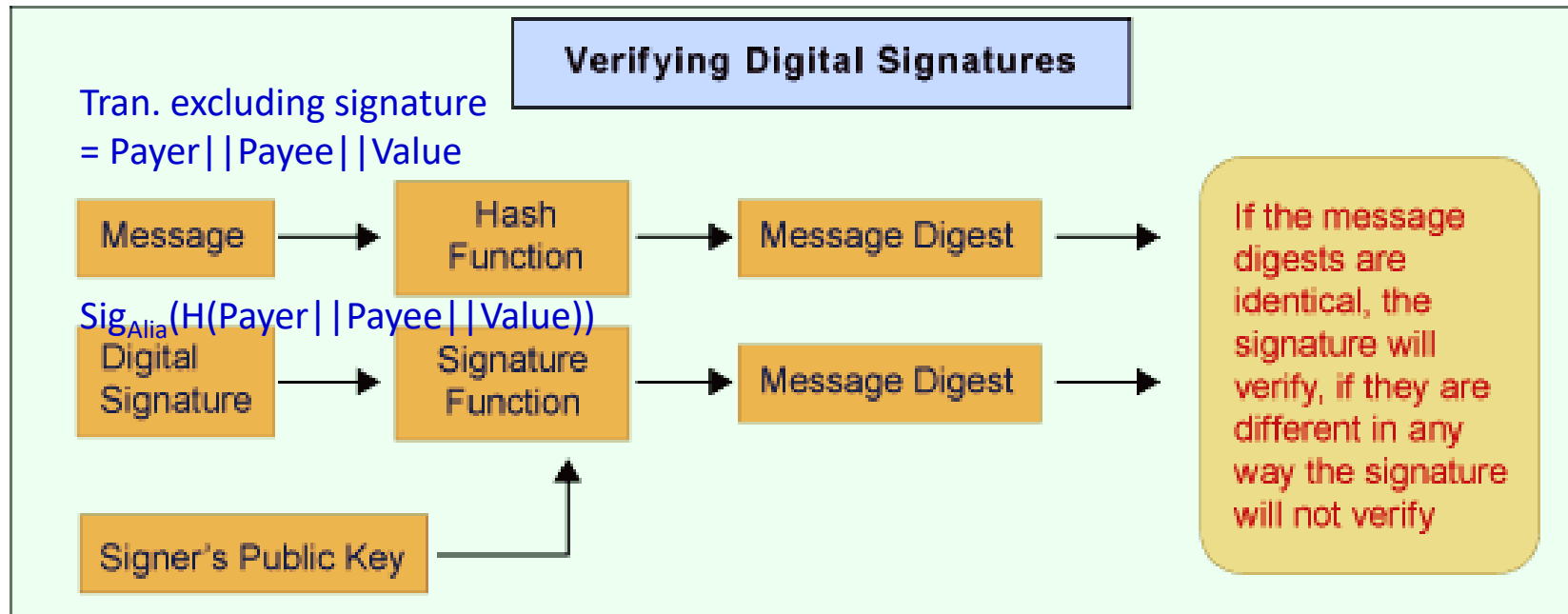
OP_EQUALVERIFY

OP_CHECKSIG

Stack

026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6a4304a5e680b2ab2cf8
304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01

Execution



Source: <https://signyourdoc.com/verify-digital-signature/>

Execution

Transaction ID:

[7cc90809ea6c90899086ed8607eea8dcc1bbe2ee3fbaa662f3f597888575c4c9](#)

ScriptSig:

<
304402205e4a6ac74d35de1ffc1188f717273bada586b3de
78f6844a548d548b0d27634c02207b127fa009b9d755b84
8772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01
> **Alia's Signatures**

<
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a4304a5e680b2ab2cf8
> **Alia's public key**
 ScriptPubKey: (Prev. Trans.)
OP_DUP
OP_HASH160
<076960c3d0cdf8076ecd0cc2200223844
75aff76e>
OP_EQUALVERIFY
OP_CHECKSIG

Stack

True



Bitcoin Transaction (legacy)

Transaction ID:

[7cc90809ea6c90899086ed8607eea8dcc1bbe2ee3fbaa662f3f597888575c4c9](#)

INPUT	OUTPUT
Prev. Trans. ID: 0f07ec1c3738873cd6336f0e2c92094bf08d0ad3bce8aba0efa2eb9d172fd60c Index: 0 ScriptSig Alia 304402205e4a6ac74d35de1ffc1188f717273bada586b3de78f6844a548d548b0d27634c02207b127fa009b9d755b848772dcee21c4a05554501fd5e0d8a60ee8b9f9d13da7d01 026a7cd9a8f65b260bd459d8c1e823110a699391cbaabb6a4304a5e680b2ab2cf8 Total Input: 0.00025302 + 0.01094000 + 0.00003616 = 0.01122918	Index 0 Value: 0.00025302 BTC scriptPubKey: OP_DUP Babar OP_HASH160 2fa9c4b9585b74fd5fd8c92fd4e32824b531e515 OP_EQUALVERIFY OP_CHECKSIG Index 1 Value: 0.01094000 BTC scriptPubKey: OP_DUP Alia OP_HASH160 9772ba84d827ac93d719a2a40f3f5dcb8963b8ad OP_EQUALVERIFY OP_CHECKSIG

Fees: 0.00003616 BTC



Bitcoin Transaction (legacy)

Transaction ID:

[7cc90809ea6c90899086ed8607eea8dcc1bbe2ee3fbaa662f3f597888575c4c9](#)

	OUTPUT
If ScriptPubKey(ScriptSig) then create output If transaction inputs are valid then	Index 0 Value: 0.00025302 BTC scriptPubKey: OP_DUP Lock to Babar's PubKey Hash OP_HASH160 2fa9c4b9585b74fd5fd8c92fd4e32824b531e515 OP_EQUALVERIFY OP_CHECKSIG Index 1 Value: 0.01094000 BTC scriptPubKey: OP_DUP Lock to Alia's PubKey Hash OP_HASH160 9772ba84d827ac93d719a2a40f3f5dcb8963b8ad OP_EQUALVERIFY OP_CHECKSIG